MILITARY SPECIFICATION

PROJECTILE, TP-T, 105MM, M468 METAL PARTS ASSEMBLY

1. SCOPE

1.1 This specification covers Projectile, TP-T, 105mm, M468, Metal Parts.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification For

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-1167	- Ammunition Data Card
MIL-STD-1168	- Lot Numbering of Ammunition
MIL-STD-1169	- Packaging, Packing and Marking of Inert Ammunition Components
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

Munitions Command

F8837835	- Pallet
F10521886	- Projectile, TP-T, 105mm, M468, Metal Parts Assembly

FSC 1315

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2.1 (Cont'd)

LI-10521886 - Inspection Equipment List for Projectile, TP-T, 105mm, M468, Metal Parts Assembly

PUBLICATIONS

Frankford Arsenal
Operating Instructions for Use of Detroit Testing Machine for Band Seat Inspection, including Drawing D8593429

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASA-B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, N. Y. 10018).

ASTM Method - American Society for Testing and Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Phila., Pa. 19103).

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and referenced specifications.

3.2 Heat treatment or stress relief.-Heat treatment or stress relief applied to any component part shall be uniform throughout and shall be so conducted as to produce the mechanical properties specified on the applicable drawing.

3.2.1 Hot forge heat treat method.-Prior to final machining, projectile bodies manufactured from forgings shall be subjected to heat treatment to attain the required mechanical properties. The heat treatment shall consist of heating above the critical temperature, not to exceed 1600 degrees fahrenheit (F), followed by oil quench. After quenching, the body forgings shall be tempered without delay at a temperature above 750 degrees (F), suitable to produce the required mechanical properties. After removal from the tempering furnace, the body forgings shall be slowly cooled in air (forced air permitted) to a temperature of not more than 700 degrees (F), afterwhich cooling may be accelerated by water spray.

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3.2.2 Cold shaping and hot cup - cold draw method.-Cold working shall be so accomplished as to produce in the projectile body the mechanical properties specified. The projectile body shall be stress relieved at a temperature of 700 degrees (F), minimum for a minimum of 30 minutes following the final forming operation during which time plastic flow takes place.

3.3 Mechanical properties.

3.3.1 Body and nose.-Mechanical properties specified by the applicable drawings shall apply.

3.3.2 Rotating bands.-After application and subsequent removal from the projectile, the rotating bands shall have the following mechanical properties:

Tensile strength 32,000 pounds per square inch (psi) min.

Elongation 15 percent per inch of gage length

3.4 Banding.

3.4.1 Band seats.-Prior to assembly of rotating band blanks, the band seats shall be clean and free from oil, grease, dirt, rust or other foreign matter. Shot or sandblasting shall not be permitted for cleaning the band seats after annealing.

3.4.2 Band blanks.-The rotating band blanks shall be in the form of a solid ring, annealed if necessary to comply with requirements of the applicable specification.

3.4.3 Application.-The rotating band blanks shall be applied after final heat treatment of the body by carefully pressing them into the band seats so that contact is made with the sidewalls for a minimum of eight-tenths of the band seat depth. There shall be no gap between band and sidewall at the outside surfaces of the body after final machining of rotating band.

3.4.4 Band seating.-The diametrical clearance between the rotating band and the band seat shall not exceed .006 inch (see 6.3).

3.5 Metal defects.

3.5.1 Components (body and nose).-The components including the cavity shall be free from cracks, splits, burst, cold shuts, pipe, porosity, inclusions, folds, seams and other metal defects.

3.5.2 Components (body and nose).-The cavity of the components shall be free of scale, fins, burrs, draw marks, laminations, imbedded foreign matter, pits and sharp edges.

3.5.3 Visual standards.-Visual standards for metal defects will be established for each contract after the first month's production. However, the establishment of visual standards will not waive the requirement for compliance with 3.5.1 and 3.5.2 (see 6.2).

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3.6 Protective coating.

3.6.1 Nose.-All aspects of surface pretreatment or surface coating are as shown by the applicable specification, subject to demonstration of control of processes through salt spray test of panels. Requirements of the component drawing apply for period of salt spray exposure without coating breakdown.

3.6.2 Body assembly.-Requirements relating to paint thickness are not specified, except as a guide for performance of tests as indicated by the applicable drawing and specifications.

3.7 Proving Ground functional tests.

3.7.1 Excess pressure.-Testing of assembled metal parts at excess pressure shall reveal no evidence of:

a. Break-up or loss of metal parts components.

b. Loose rotating band(s) attributable to noncompliance with the drawing or the specification requirements.

c. Shear of rotating band(s) as exemplified by complete lack of engraving.

3.8 Workmanship.-The requirements for workmanship are as shown on the applicable drawings, in the applicable specifications and the following:

3.8.1 Threads.-Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.8.2 Burr.-No part shall have a burr which might interfere with the assembly or function of the round or which might be injurious to personnel handling the item.

3.8.3 Foreign matter.-All parts and assemblies shall be free of chips, dirt, grease, rust, corrosion or other foreign matter.

3.8.4 Rotating band damage.-The rotating band after machining shall be protected from being nicked, burred or otherwise damaged during subsequent handling and preparation for shipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Quality Assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection criteria.

a. Successful passage of all inspections required by the drawings and specifications shall be considered as a condition precedent to performing the Proving Ground functional test (see 3.7).

b. Acceptance or rejection of lot by the Government shall be contingent on successful passage of the Proving Ground function tests in 4.3.2.

4.2 First article sample.

4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 35 metal parts assemblies as follows:

25 metal parts assemblies complete (painted)
5 metal parts assemblies complete (unpainted)
5 sets of component parts unassembled and unpainted including rotating band blanks (bodies to be completely machined, including band seat)

Twenty of the above painted metal parts assemblies shall be forwarded to a designated Proving Ground for ballistic testing. The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply. A first article sample or portion thereof, as directed by the procuring agency shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing processes, materials used, drawings, specifications or source of supply, such as to significantly affect product uniformity as determined by the responsible technical agency.

4.2.1.1 Examination and test.-After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.-Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions and test methods and procedures.

4.3.1 Lot.

4.3.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Mechanical properties.-(see 6.9)

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4.3.1.2.1 Hardness and tension test.

4.3.1.2.1.1 Heat treatment lots (definition).-Heat treatment lots shall contain heat treated components from not more than one heat of steel that have been heat treated in the same heat treatment equipment, at the same temperature, in one unchanged process. The heat treatment lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.3.1.2.1.2 Hardness (heat treatment lots).-In order to determine uniformity of heat treatment, each component in the heat treatment lots (as defined by 4.3.1.2.1.1) shall be subjected to a Brinnell hardness test or any method of hardness testing approved by the procuring activity.

4.3.1.2.1.3 Heat treatment lots (tension test samples).-Two components, one of the hardest and one of the softest as determined by the hardness test shall be subjected to this test. Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot. At the option of the supplier, a production control sample may be utilized to establish a hardness range. When a production control sample is used, the hardness range thus established shall be maintained throughout the heat treatment lot. The size of the sample shall be determined by the contractor. Components which fall outside of the established range may be re-heat treated to fall within the range or new mechanical properties may be determined from newly selected projectile components which will represent the true extremes of hardness in the heat treatment lot.

4.3.1.2.1.4 Stress relieved lots (definition).-Stress relieved lots shall contain stress relieved components, from not more than one heat of steel, that have been cold shaped by the same process, utilizing the same tooling and equipment and stress relieved in the same furnace at the same temperature for the same length of time. The stress relieved lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.3.1.2.1.5 Hardness (stress-relieved lots).-Twenty-five cold shaped components from the first batch of each stress-relieved lot (as defined by 4.3.1.2.1.4) processed through the stress-relief furnace and five components from each one hour's production thereafter for the balance of the stress relieved lot shall be subjected to a Brinnel hardness test, or any method of hardness testing approved by the procuring activity to assess uniformity of the product.

4.3.1.2.1.6 Stress relieved lots (tension test samples).-Two components, one of the hardest and one of the softest, as determined by the hardness test from the first twenty-five components shall be subjected to this test. Failure of any tension test specimen to comply with the mechanical properties requirement of the applicable drawing shall be cause for rejection of the lot. When it has been determined that the mechanical properties comply and furnace temperature and time have

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4.3.1.2.1.6 (Cont'd)

been established, the furnace temperature and time shall be maintained throughout each stress-relief lot. The furnace temperature may vary within twenty-five degrees of Fahrenheit but not below the minimum temperature of 700 degrees Fahrenheit and the minimum 30 minutes soak time. Furnace temperature and time will be recorded every hour to assess adequate control of stress relief equipment. The furnace temperature will be calibrated each eight hours by an independent method.

4.3.1.2.1.7 Miscellaneous lots (definition).-Small lots (such as components manufactured from abnormally small heats of steel or bodies that have become separated from previously delivered lots, etc.) may at the discretion of the supplier be grouped into miscellaneous heat treatment or stress-relieved lots as the case may be. Such lots shall be identified by a miscellaneous lot code number and processed in accordance with either heat treat or stress-relieved lots as applicable. The miscellaneous lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.3.1.2.1.8 Hardness (miscellaneous lots).-Miscellaneous heat treatment lots shall be subjected to a hardness test in accordance with 4.3.1.2.1.2. Miscellaneous stress-relieved lots shall be subjected to a hardness test in accordance with 4.3.3.2.1.5.

4.3.1.2.1.9 Miscellaneous lot (tension test samples).-Samples for tension test of miscellaneous heat treatment or stress-relieved lots shall be selected in accordance with 4.3.1.2.1.3 or 4.3.1.2.1.6, as applicable. Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot. Tension tests shall be performed on each miscellaneous heat of steel before being grouped into a miscellaneous lot.

4.3.1.2.1.10 Hardness test methods.-After final heat treatment or stress relief, but prior to finish machining, a Brinnel hardness test shall be performed on the outside surface at the band seat location or at any other area on the body deemed necessary for process control. The equipment and procedures for performing this test are prescribed by ASTM Method E-10, when a Brinnel hardness test is used.

4.3.1.2.1.11 Tension test methods.-From the samples provided by the heat treat, stress relief or miscellaneous lots two tension test specimens conforming to the largest obtainable standard "R" type specimen, as prescribed by ASTM Method E-8, shall be taken longitudinally from opposite sides of the body at midwall, in such a manner that the center of the gage length of the machined test bar will fall in line coincident with the center of the rotating band seat. The number of specimens and their location is to determine the acceptability of the mechanical properties requirements and with the approval of the procuring agency, the Government shall not be restricted from testing additional longitudinal specimens at other locations to determine that the mechanical properties of the projectile body meet the drawing

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4.3.1.2.1.11 (Cont'd)

requirements. The criteria for performing this test shall be prescribed by ASTM Method E-8; except that elongation shall be measured to the nearest thousandth (0.000) prior to conversion into percent elongation. Percent elongation shall be recorded to the nearest tenth (0.0 percent).

4.3.1.2.1.12 Nose (tension test samples).-Two noses from each lot of 1000 or fraction thereof shall be subjected to the tension test. Failure of any specimen to comply with the yield strength or elongation requirement of the applicable drawing shall be cause for rejection of the lot represented by the sample. The test may be performed prior to finish machining.

4.3.1.2.1.13 Nose (tension test method).-From the samples supplied above one tension test specimen conforming to the largest obtainable standard "R" type specimen shall be taken longitudinally from each component. The criteria for performing this test shall be as prescribed by ASTM Method E-8.

4.3.1.2.1.14 Projectile lots.-When it has been determined that components of a heat treated, stress relieved or miscellaneous lot have complied with the mechanical properties requirement of the applicable drawing, they may be regrouped for assembly into projectile lots, to which will be assigned a projectile lot number in accordance with MIL-STD-1168. The projectile lot shall consist of projectile assemblies produced by one supplier under one contract in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision, under conditions of continuous production, free from interruptions other than those due to the end of a shift, day, or work week. The projectile lot thus constituted shall be the basis for the selection of samples for the functional test hereinafter prescribed.

4.3.1.2.1.15 Rotating band (tension test samples.)-One body assembly shall be selected from each 8-hour production or fraction thereof from each banding machine in operation and subjected to the tension test to determine compliance with 3.3.2. Failure to comply shall be cause for rejection of the production represented by the sample.

4.3.1.2.1.16 Rotating band (tension test method).-The samples provided above shall have the rotating band removed and carefully flattened without hammering so as to hold to a minimum any additional cold working. The specimen blank shall be machined so that all serration marks from the band seat are removed. A test specimen having one-inch gage length and a one-half inch gage width, and grip ends of sufficient length to accomodate the tensile machine shall be removed from the blank and tested in accordance with thy procedures prescribed by ASTM Method E-8.

4.3.1.3 Band seating.

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4.3.1.3.1 Tightness.

4.3.1.3.1.1 Non-destructive (samples).-Five body assemblies from each 1-hour production or fraction thereof per banding machine in operation shall be subjected to this test. Failure of one or more bands to comply with 3.4.4 shall be cause for rejection of the production represented by the sample.

4.3.1.3.1.2 Non-destructive method.-The samples provided by 4.3.3.3.1.1 shall be tested utilizing the equipment and procedures prescribed by Frankford Arsenal Detroit Testing Machine Manual (see 6.3).

4.3.1.3.1.3 Destructive (samples).-When the non-destructive equipment is not available, one body assembly from each 8-hour production or fraction thereof from each banding machine in operation shall be subjected to this test to determine compliance with 3.4.4. Failure to comply shall be cause for rejection of the production represented by the sample.

4.3.1.3.1.4 Destructive method.-The body assembly selected above should not be processed through the band machining line, although each band should have a light surface cut to true up the surface prior to measuring. The diameter of each band shall be measured on 3 locations, 60 degrees apart, on the points along the cylindrical portion of their length, near the front and rear edges. This comprises 6 diametrical measurements per band. These points of measurement shall be marked on the bands and on the projectile body adjacent thereto in such a manner as to permit proper body and band orientation after removal. The bands shall be carefully removed so as to keep deformation to a minimum. After removal, the thickness of the band shall be measured at each of the marked points (a total of 12 measurements) and the diameter of the band seat corresponding to these points shall also be measured. Add to each of the diameters of the band seat the sum of the corresponding 2 thicknesses of the band; the results obtained shall then be subtracted from the previously determined diameters over the band.

4.3.1.3.1.5 Sidewall contact.-The samples provided for the tension test shall be sectioned by cutting through two planes normal to the longitudinal axis one-half inch forward and rear of rotating band. The cylindrical portion containing the rotating band shall be sectioned approximately in half through a plane parallel to its longitudinal axis. One of the two halves shall be used for establishment of mechanical properties of the band; the other half shall be sectioned into two parts, thus exposing four surfaces each of which reveals a veiw of the band swaged into the band seat. The exposed surface shall be subjected to a light grinding operation to remove metal smear effects of saw cuts prior to measurements for depth of sidewall contact to determine compliance with the requirements of 3.4.3.

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4.3.1.4 Cleaning and application of protective coating.-The sample size and acceptance criteria for the inspection of cleaning the item and the application of protective coating shall be as specified in the applicable specification. Failure to comply with the requirements shall be cause for taking the action recommended by the applicable specification.

4.3.1.4.1 Salt spray.-The sample size and acceptance criteria for salt spray testing shall be in accordance with the applicable specification. Failure to comply with the requirements of the applicable drawing shall be cause for taking corrective action recommended by the applicable specification. The salt spray test prescribed by the applicable drawing shall be performed utilizing the equipment and procedures specified by ASTM B-117.

4.3.1.4.2 Protective coating.-The procedures for testing the cleaning and surface treatment methods for components and assemblies shall be in accordance with the methods and criteria for testing as specified in the applicable specification.

4.3.2 Proving Ground functional tests.-The following Proving Ground tests will be performed by the Government.

4.3.2.1 First Article sample.

4.3.2.1.1 Initial production sample.-The quantities listed in Table I shall be ballistically tested in the phases indicated.

TABLE I

Phase	Quantity
Excess pressure	10
Service pressure	10

4.3.2.2 Production lot sample.-The quantities listed in Table II shall be ballistically tested in the phases indicated.

TABLE II

Phase	Quantity
*Excess pressure	10

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*After three (3) consecutive production lots have been tested successfully the excess pressure phase may be discontinued, however, implementation of the excess pressure phase shall be instituted whenever directed by the procuring agency.

4.3.2.3 Excess pressure.-The projectiles supplied in accordance with 4.3.2, will be fired for recovery, in a gun for which the projectile is standard, the tube of which will have 50 percent life remaining, based on current gun tube condemnation criteria. A propellant charge adjusted to produce a pressure of 52,000 PSI to 56,000 PSI shall be used. Photographic equipment will be set up near the gun muzzle to record the flight of the projectile, for the initial production sample and the first three (3) production lot samples fired. An attempt will be made to recover at least 60 percent of the projectiles fired in this phase for examination to determine compliance with 3.7. Failure to comply shall be cause for rejection of the initial production sample or the production lot represented by the sample.

4.3.2.4 Service pressure.-The projectiles supplied in accordance with 4.3.2 will be fired alternately with projectiles from a reference lot for comparison. The projectile shall be fired in a gun for which the round is standard, the tube of which shall have a minimum remaining service life of 50 percent, based upon current gun tube condemnation criteria. The projectile will be fired for accuracy at a range selected by the proof director. Photographic equipment will be set up near the gun muzzle to record the flight of the projectile to determine compliance with 3.7. Failure to comply shall be cause for rejection of the initial production sample represented by the sample.

4.3.3 Inspection equipment.-Index of Inspection Equipment List No. LI-10521886 identifies the inspection equipment required to perform the examination and tests prescribed herein. Reference is made to the LIfor definition of coding format.

4.3.4 Examination.-One-hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105. An AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect shall be used. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual inspection. The roughness comparison specimens prescribed by ASA-B46.1 shall be used as a basis for surface roughness determination.

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4.3.4.1 Tool control dimensions.-Dimensions marked tool control shall be gaged during first article inspection and after each 250 items produced for each machine in operation. Where destruction of components is necessary to inspect these dimensions, measurement of the tool may be substituted providing the contractor has established correlation between tool dimensions and component dimensions prior to start of production. The contractor shall be responsible for maintaining adequate surveillance to assure drawing requirements are being met throughout the life of the tool. Corrective action will immediately be instituted by the contractor whenever a defective tool control dimension is found.

4.3.4.2 Nose, Drawing D10521889, covering a detail of Drawing F10521886.

Defect		Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Runout of cavity diameter with pitch diameter of thread	Gage
102.	Pitch diameter of thread, min.	Gage
103.	Major diameter of thread, min.	Gage
104.	Diameter of shoulder	Gage
105.	Diameter of cavity for min. depth	Gage
106.	Depth to bottom of cavity	Gage
107.	Length from rear face to inside basic diameter	Gage
108.	Length from front face to shoulder	Gage
109.	Length, total	Gage
110.	Profile of ogive radius	Gage
111.	Metal defective	Visual (Probe light)
MINOR		
201.	Diameter of thread undercut	Gage
202.	Width of thread undercut, max.	Gage
203.	Surface finish improper	Visual (see 6.1)
204.	Radius or chamfers missing or incorrect	Visual (see 6.1)
205.	Burr	Visual
206.	Foreign matter	Visual

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4.3.4.3 Body Assembly, Drawing F10521838, (prior to handling) covering a detail of Drawing F10521886.

	Defect	Method of Inspection
MAJOR		
101.	Diameter of forward rotating band seat	Gage
102.	Diameter of rear rotating band seat	Gage
103.	Width of forward rotating band seat	Gage
104.	Width of rear rotating band seat	Gage
105.	Location of forward rotating band seat	Gage
106.	Location of rear rotating band seat	Gage
107.	Scoring missing from rotating band ribs	Visual
108.	Metal defective	Visual (Probe light)
MINOR		
201.	Foreign matter on rotating band seat	Gage

4.3.4.4 Body Assembly (prior to paint) Drawing F10521887, covering a detail of Drawing F10521886.

CRITICAL		
	None defined	
MAJOR		
101.	Runout of pitch diameter of forward thread with outside body diameter and front face	Gage
102.	Runout of pitch diameter of rear thread with outside body diameter and rear face	Gage
103.	Runout of forward inside diameter of cavity with outside body diameter	Gage
104.	Runout of rear inside diameter of cavity with outside body diameter	Gage
105.	Runout of forward rotating band flats with outside body diameter	Gage
106.	Runout of rear rotating band flat with outside body diameter	Gage
107.	Pitch diameter of forward thread, max.	Gage
108.	Minor diameter of forward thread, max.	Gage
109.	Diameter of front face	Gage
110.	Diameter of cavity (forward)	Gage
111.	Diameter of cavity (rear)	Gage
112.	Diameter of forward rotating band flats	Gage

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Defect	Method of Inspection
113. Diameter of rear rotating band flat	Gage
114. Diameter of boss	Gage
115. Length of effective thread, min. (forward)	Gage
116. Length of effective thread, min. (rear)	Gage
117. Length of forward thread, max.	Gage
118. Length of rear thread, max.	Gage
119. Length of boss	Gage
120. Length, total	Gage
121. Location of forward rotating band	Gage
122. Location of rear rotating band	Gage
123. Thickness of base	Gage
124. Width of forward rotating band	Gage
125. Width of rear rotating band	Gage
126. Metal defective	Visual (Probe light)

MINOR

201. Depth of undercuts front and rear of rotating bands (if applicable)	Visual (see 6.1)
202. Width of undercuts front and rear of rotating bands (if applicable)	Visual (see 6.1)
203. Radii or chamfers missing or incorrect	Visual (see 6.1)
204. Surface finish improper	Visual (see 6.1)
205. Burr	Visual
206. Foreign matter	Visual

4.3.4.5 Metal Parts Assembly (after paint) Drawing F10521886.

CRITICAL

None defined

MAJOR

101. Weight	Approved Scale
102. Runout of nose with outside body diameter	Gage
103. Pitch diameter of rear thread, max.	Gage
104. Minor diameter of rear thread, max.	Gage
105. Pitch diameter of rear thread, min.	Gage
106. Length, total, max.	Gage
107. Diameter of front face greater than nose diameter on circumference	Visual
108. Metal defective	Visual (Probe light)

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4.3.4.5 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Protective coating inadequate or damage	Visual
202. Paint on rotating band bearing surfaces	Visual
203. Rotating band(s) damaged	Visual
204. Marking missing incorrect or illegible	Visual
205. Foreign matter	Visual

4.3.4.5 Pallet.

CRITICAL

None defined

MAJOR

None defined

MINOR

201. Strapping, missing base or broken	Manual/Visual
202. Pallet damaged	Visual
203. Improperly packed	Visual
204. Marking, incorrect or illegible	Visual

4.3.5 Body diameter gage (before paint).-Each projectile body in the lot shall be gaged in a Government certified cylindrical body diameter gage. Failure of the projectile body to comply with the requirements of the applicable drawing shall be cause for rejection of the projectile body.

4.3.6 Body diameter gage (after paint).-Each projectile in the lot shall be gaged in a Government certified cylindrical body diameter gage. Failure of the projectile to comply with the requirements of the applicable drawing shall be cause for rejection of the projectile.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking, Level C.-Packing and marking shall be in accordance with MIL-STD-1169 and this specification.

5.2 Packing.

5.2.1 Body assembly.-Body assembly, part number 10521886, shall be packaged in accordance with Drawing F8837835.

5.3 Marking.

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5.3.1 Body assembly.-Palletized body assemblies shall be marked in accordance with Drawing F8837835.

5.4 Shipping.-When component parts from more than one lot are shipped as a carload, each lot must be kept separate and divisions between lots clearly indicated to prevent mixing in transit.

5.5 Data cards.-Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Visual examination qualification.-When compliance with the applicable requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.2 Visual standards.-Visual standards to be established at contractor's plant in compliance with 3.5.3.

6.3 Band seating.-The operating procedures for use of Detroit Testing Machine as prepared by the Quality Branch, US Army, Frankford Arsenal, SMUFA-Q3200.

6.4 It is the responsibility of the supplier of the projectile metal parts, to select a fully-killed carbon steel of a grade compatible with the abilities of the equipment and the methods of processing the metal within the operations of the shops of forming and machining, to produce the shell to the dimensions and properties specified as an end item requirement.

6.5 The supplier of the selected steel raw material is responsible only for furnishing a metal composition grade and type with dimensions, weight, soundness and freedom from metallurgical or mechanical defect limitations, identifiable with the standard AISI grade selected by the projectile manufacturer and complying with Specification ASTM, A576 special bar quality.

6.6 In the application of carbon steel, blooms, billets and slabs, forging quality processing techniques adapted to use of selected steel should be applied with the expectation that there will be variables in the metal cross-sectional dimensions in weight per unit length in the hardenability variance from heat to heat, in the mechanical condition of the surface and in the straightness of the raw material. Economic advantages will be realized if the process and equipment within the shops of the forger and the machiner have been engineered in terms of equipment sequence of operations, tooling designs, handling devices and processing techniques having capabilities of compensating for such variables.

6.7 Heat treatment is understood to be any thermal process which brings about a modification in the mechanical or metallurgical properties of the metal involved.

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6.8 Ordering data.-Procurement documents should specify the title, number and date of this specification.

6.9 Alternative mechanical properties testing.-Alternative procedures for mechanical properties testing may be used by the contractor when such testing and procedures provide as a minimum the quality assurance specified herein. Prior to applying the alternative procedures, the contractor shall describe them in a written proposal and shall demonstrate for the approval of the responsible technical agency that their effectiveness is equal to or better than the quality assurance provisions specified herein.

Custodian:

Army - MU

Preparing activity:

Army - MU(FA)

Project no. 1315-A238

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Department of the Army

OFFICIAL BUSINESS

Commanding Officer
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Philadelphia, Pa. 19137

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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